



SIH Livelihood Navigator

NBA Inspection Defense & Project Presentation Guide

Comprehensive, simple, and technical walkthrough for evaluation.

Ready for Mobile & Desktop Viewing



1. Executive Summary (60-Second Pitch)

"What are we building from scratch?"

We built the **SIH Livelihood Navigator**, an end-to-end, AI-powered digital ecosystem designed to empower rural and underprivileged citizens by matching them with targeted skill training (**NSQF - National Skills Qualifications Framework**), government livelihood schemes, local jobs, and self-employment opportunities.



Real-World Problem We Solve

- **Information & Language Barrier:** Rural job seekers struggle with complex, formal scheme documents.
- **Skill Mismatch:** Youth don't know which NSQF courses fit their education, age, and interest.
- **Lack of Tracking:** Government officers lack visibility into candidate progress and employment outcomes.



Our Solution

-  **AI Voice Assistant:** Multilingual speech/text interaction for semi-literate users.
-  **Smart Recommendation Engine:** Suitability scoring matching profiles to NSQF courses.
-  **Visual Career Roadmap:** Step-by-step guidance from training to employment.
-  **Nodal Officer Dashboard:** Allows officers to monitor candidates, log interventions, and track outcomes.



2. Real-World Case Study (Story Explanation)



Meet Ramesh (22 years old, 10th Pass, Rural MP)

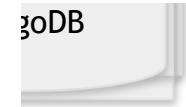
1. **Voice Entry:** Ramesh opens the app and speaks in Hindi: "*Mujhe Mobile Repairing seekhna hai*".
2. **AI Recommendation:** System processes his age (22) and education (10th pass), recommending **NSQF Level 4 - Field Technician** at a nearby PMKVY center and Mudra loan details.
3. **Visual Roadmap:** Ramesh gets a 5-step roadmap: Registration → Training → Certification → Apprenticeship → Job.
4. **Officer Support:** A Nodal Officer tracks Ramesh in the dashboard and logs a financial toolkit intervention.
5. **Outcome:** Ramesh earns certification and lands a job paying ₹18,000/month.



3. High-Level System Architecture

Our project uses a decoupled 3-tier architecture (Client, Server API, AI Microservice, Database):

goDB



4. Technical Data Workflow

sequenceDiagram actor User as Beneficiary / Officer participant FE as React Frontend participant API as Express API Server participant REC as Recommendation Engine participant DB as MongoDB Database User->>FE: Inputs Profile (Age, Education, Location, Goal) FE->>API: POST /api/beneficiaries/profile (Bearer JWT) API->>DB: Save/Update Beneficiary Record API->>REC: Calculate Recommendation Scores REC->>DB: Query NSQF Courses & Opportunities DB-->>REC: Return Matching Items REC-->>API: Ranked List with Match Score % API-->>FE: HTTP 200 OK (JSON Response) FE-->>User: Render Recommendation Feed & Visual Roadmap

 5. Recommendation Algorithm Mechanics

Match scores (0-100%) are calculated dynamically using the formula:

Total Score = EducationMatch (30%) + AgeEligibility (20%) + LocationProximity (20%) + InterestAlignment (30%) - BarrierPenalties

- **Education Matching (30%):** Verifies minimum education requirement (e.g., 10th pass for Level 4).
- **Age Eligibility (20%):** Validates scheme age restrictions (e.g., 18-35 years).
- **Location Proximity (20%):** Prioritizes district & state level training centers.
- **Interest Alignment (30%):** Keyword vector matching between candidate goal & course content.

 6. Technology Stack Overview

Layer	Technology	Role in Project
Frontend	React.js, Vite, Tailwind CSS	Responsive mobile UI, dynamic charts & visual roadmaps.
Backend API	Node.js, Express.js	REST APIs, authentication, request validation.
Database	MongoDB, Mongoose ODM	NoSQL database for profiles, courses, enrollments & outcomes.

Layer	Technology	Role in Project
Security	JWT, Bcrypt	Password hashing & token authorization.
AI Engine	Python (FastAPI), Gemini API	Speech-to-text, NLP parsing & query understanding.
Testing	Node Test Runner, Supertest	Automated backend API testing.

 7. Database Entity Relationship Diagram

erDiagram
 USER ||--o{ BENEFICIARY_PROFILE : "has profile"
 USER ||--o{ INTERVENTION : "officer logs"
 BENEFICIARY_PROFILE ||--o{ TRAINING_ENROLLMENT : "enrolls in"
 NSQF_COURSE ||--o{ TRAINING_ENROLLMENT : "contains"
 BENEFICIARY_PROFILE ||--o{ OUTCOME : "achieves"
 USER {
 string name
 string email
 string role
 }
 BENEFICIARY_PROFILE {
 number age
 string educationLevel
 string district
 array skills
 }
 NSQF_COURSE {
 string title
 number nsqfLevel
 string minEducation
 }
 INTERVENTION {
 string type
 string notes
 }
 OUTCOME {
 string status
 number monthlyIncome
 }

 8. Software Engineering Methodology

- **Modular 3-Tier Architecture:** Decoupled frontend, REST backend, and AI engine for independent scalability.
- **Security First:** Salted password encryption (`bcrypt`), JWT token validation, and input sanitization (`express-validator`).
- **Optimized Queries:** Mongoose schemas with indexed fields (`district`, `nsqfLevel`).
- **Automated Testing:** Integration test suites (`tests/api.test.js`) executed with in-memory MongoDB databases.

 9. NBA Defense Questions & Answers

Q1: How is this different from Naukri or LinkedIn?

Commercial portals target white-collar English speakers with resumes. We target the unorganized rural sector with voice support in local languages, NSQF accreditation alignment, and government officer intervention.

Q2: Why is NSQF integration important?

NSQF standardizes skill qualifications from Level 1-10 across India, ensuring certificates are nationally recognized for jobs and government loans (PMKVY / Mudra).

Q3: What value does the Officer Dashboard provide?

It solves the last-mile monitoring problem. Nodal officers can track drop-outs, provide financial/equipment aid, and measure scheme employment success.

**10. Tomorrow's Prototype Demo Checklist**

1. **Beneficiary Registration:** Input age (22), education (10th Pass), district, and interest.
2. **Recommendation Feed:** Demonstrate match score % generation.
3. **Voice Query Demo:** Speak/type natural language query.
4. **Visual Skill Roadmap:** Show step-by-step career path.
5. **Officer Dashboard:** Log an intervention and view district outcome metrics.